

## Q.16 — Symmetry of Four Three-Fold Axes

DMG 2019 · Mineralogy: Crystal Morphology & Symmetry · Reference: Perkins, Mineralogy, 3rd ed.

**Q.16.** Four three-fold axes, each inclined at  $54^{\circ}44'$  to the crystallographic axes, are characteristic symmetry of which one of the following crystal systems?

- (a) Hexagonal
- (b) Isometric
- (c) Orthorhombic
- (d) Triclinic

### 1. The Underlying Concept — How Rotation Axes Combine

Perkins explains that symmetry axes in crystals cannot combine in random ways — only **six** geometrically valid combinations of rotation axes exist. Four are the simple combinations 222, 32, 422, and 622. The remaining two combinations are built entirely around the geometry of a **cube**, and these are the combinations that define cubic (isometric) symmetry.

*"The other two (Figures 18e and f) are best envisioned by thinking about a cube. The angles between rotation axes are all special angles (to avoid infinite symmetry). In Figures 18e and f, the angles are equal to angles between various diagonals in a cube and perpendiculars to cube faces."*

— Perkins, *Mineralogy*, p. 251

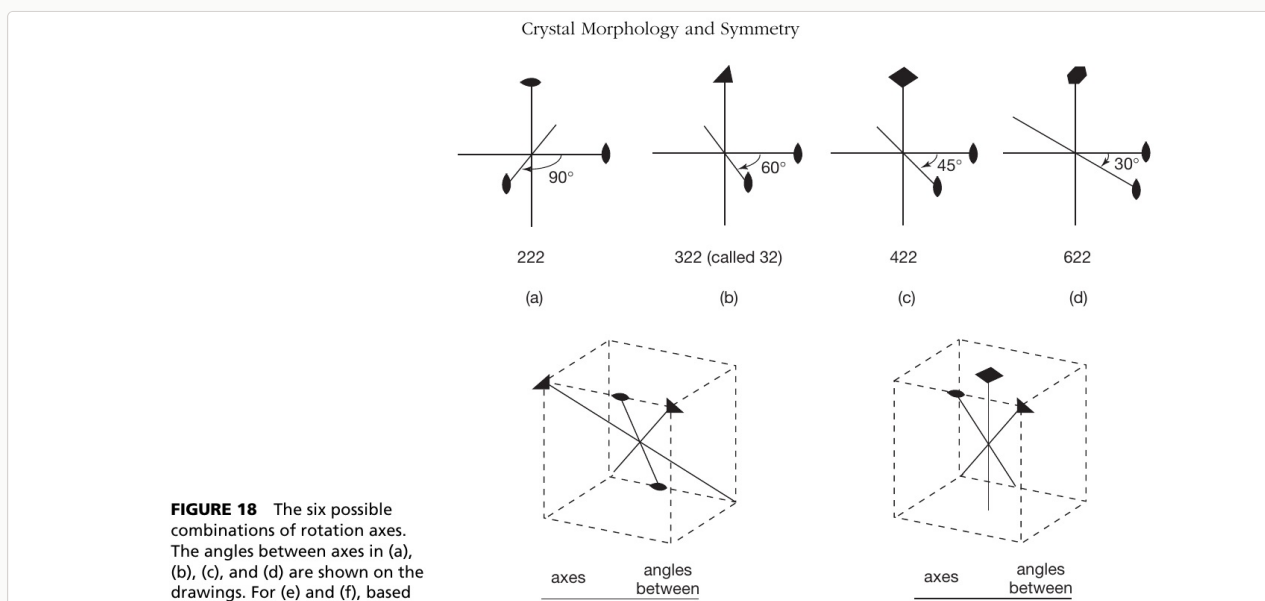


Figure 18 (Perkins, *Mineralogy*, p. 251) — The six possible combinations of rotation axes. Combinations (e) and (f) are based on a cube; their angle tables list  $70^{\circ}32'$ ,  $54^{\circ}44'$ ,  $45^{\circ}00'$ , and  $35^{\circ}16'$  — the special angles between a cube's edges, face diagonals, and body diagonals.

### Where $54^{\circ}44'$ comes from — the geometry

The three crystallographic axes of a cubic crystal run along the cube's **edges** (coinciding with the three 4-fold axes). The **body diagonals** of the cube are the **four 3-fold axes**. The angle between an edge and a body diagonal of a cube is fixed by simple solid geometry:

$$\theta = \arccos(1/\sqrt{3}) = 54.7356^{\circ} = 54^{\circ}44'$$

Because a cube has exactly **four** space diagonals, this construction always produces **four** 3-fold axes, each tilted  $54^{\circ}44'$  from all three crystallographic axes — never more, never fewer, and only in a cubic arrangement.

### 2. Confirming the Diagnostic Symmetry of the Cube

"Cubes have the highest symmetry possible for crystals: three 4-fold axes, **four 3-fold axes**, six 2-fold axes, nine mirror planes, and an inversion center."

— Perkins, *Mineralogy*, p. 244

"**Cubic System:** This system is also called the **isometric system**. Crystals have high symmetry, all having **four 3-fold or 3 axes**. Some have three 4-fold axes as well, and some have 2-fold axes or mirror planes."

— Perkins, *Mineralogy*, p. 261

The book even sets this exact relationship as a self-test exercise at the end of the chapter:

"All crystals with **four 3-fold axes of symmetry must belong to the cubic system**."

— Perkins, *Mineralogy*, p. 302 (Questions for Thought, Q.3)

So the presence of exactly **four** three-fold axes — not one, not two — is the single diagnostic fingerprint of the isometric system, and their fixed 54°44' inclination to the crystallographic axes is the direct geometric fingerprint of a cube's body diagonals.

### 3. Evaluating Each Option

| Option               | Symmetry Content                                                                                                               | Verdict        |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------|
| (a) Hexagonal        | Only <b>one</b> 3-fold, 3, or 6-fold axis. Perkins: "Crystals with more than one 3-fold or 3 axis belong to the cubic system." | Eliminated     |
| <b>(b) Isometric</b> | <b>Four</b> 3-fold axes, each inclined at 54°44' to the three crystallographic (edge) axes — exact match to the question       | <b>Correct</b> |
| (c) Orthorhombic     | Three mutually perpendicular <b>2-fold</b> axes only; no 3-fold axes at all                                                    | Eliminated     |
| (d) Triclinic        | Lowest symmetry class — at most a center of inversion; no rotation axes higher than 1-fold                                     | Eliminated     |

✓ Answer: (b) Isometric

### 4. Probable Follow-Up Exam Questions

**Q. What is the total symmetry content of the highest-symmetry cubic point group (m3m, hexoctahedral class)?**

A. Three 4-fold axes, four 3-fold axes, six 2-fold axes, nine mirror planes, and a center of symmetry.

**Q. What is the general (48-face) form of the isometric system called?**

A. The hexoctahedron.

**Q. Why are isometric-system minerals optically isotropic?**

A. The four 3-fold axes make atomic arrangement equivalent in every direction, so there is no unique optical direction — refractive index is identical in all directions.

**Q. What angle does a cube's face diagonal make with its edge (relevant to the 4-2 axis combination in Figure 18f)?**

A. 45°00'.

**Q. Which crystal system is optically uniaxial, with its optic axis parallel to a single 3-fold, 3, or 6-fold axis?**

A. Hexagonal system.